Design and Implementation of a Digital pH Meter with Analog Conditioning Circuit Calibration

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***Abstract*— This paper introduces a novel digital pH meter designed to enhance the accuracy, portability, and ease of use in pH measurement. The proposed device combines modern digital technology with traditional pH sensing methods to overcome the limitations of conventional analog conditioning circuits and their calibration. The study begins with a review of existing pH measurement techniques and devices, which informs the development of a microcontroller-based prototype. Extensive testing demonstrates that the device offers accuracy and reliability on par with laboratory-grade instruments. Notable features include portability, real-time data logging, automatic temperature compensation, and calibration reminders. This digital pH meter provides a cost-effective, versatile solution for diverse applications, marking a significant step forward in pH measurement technology.**

Keywords—PH meter, ESP32 microcontroller, digital, LCD display

I. Introduction

pH measurement plays a crucial role in many scientific and industrial fields, such as environmental monitoring, chemical analysis, water treatment, and food processing. Accurate pH measurement is essential to ensure product quality, compliance with regulatory standards, and safe operational conditions. Traditionally, pH meters have been analog, but recent advancements in digital technology have led to the development of digital pH meters that offer improved precision, ease of use, and additional functionality. These devices incorporate microprocessors, advanced sensors, and digital displays, making them more reliable and accessible for a range of applications.

The measurement and control of pH, which represents the hydrogen ion concentration or acidity/basicity of a solution [1], is of paramount importance across numerous scientific disciplines and industrial applications.The pH scale is defined by the negative logarithm of the hydrogen ion activity or effective concentration [2]. A pH of 7 is considered neutral, with solutions having a pH less than 7 being acidic, and solutions with a pH greater than 7 being basic or alkaline. The lower the pH, the higher the acidity; the higher the pH, the higher the alkalinity [3]. The pH value provides crucial information about the chemical nature and reactivity of a solution. It governs various chemical and biological processes, such as enzyme activity, protein structure, and solubility of molecules. Even a slight change in pH can significantly impact the properties and behaviors of substances in solution. Historically, the developmentof pH measurement has been a pivotal advancement in the fields of chemistry, biochemistry, and related areas. It was in 1909 that Danish chemist Søren Peder Lauritz Sørensen introduced the concept of the pH scale [4], which quantifies the acidity or basicity of a solution based on the negative logarithm of the hydrogen ion concentration. This groundbreaking work built upon earlier theories by pioneers such as Arrhenius, Brønsted, and Lowry, who established the foundations of acid-base chemistry Traditional methods for pH measurement, such as litmus paper and colorimetric indicators, have been widely used but often lack the desired accuracy, precision, and convenience for many applications [5]. The advent of electrochemical pH meters, which employ specialized electrodes and electronic circuitry, revolutionized the field by providing rapid, reliable, and quantitative pH measurements. However, commercial pH meters can be costly, and their performance can be influenced by factors such as temperature, ionic strength, and electrode aging [6,7].

This paper explores the design and recent innovations in digital pH meters, focusing on advancements in sensor technology, signal processing, and user interface design. We aim to provide an overview of the key considerations in the design of these devices, along with a discussion of their impact on various scientific and industrial applications..

II. PH Measurement, Theoretical Concepts

A. Definition of pH

pH is a measure of the acidity or basicity of an aqueous solution. It represents the negative logarithm of the hydrogen ion concentration (H+) in the solution. The pH scale ranges from 0 to 14, with a value of 7 representing a neutral solution, values below 7 indicating an acidic solution, and values above 7 indicating a basic (alkaline) solution [1].

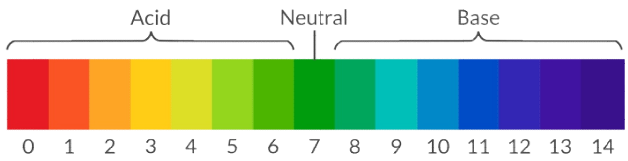

Fig. 1. PH scale [8]

A major breakthrough in pH measurement occurred in 1934 with the introduction of the glass electrode by American chemist Arnold Beckman [9]. The glass electrode, consisting of a thin glass membrane selectively permeable to hydrogen ions, revolutionized pH measurement by allowing for direct and precise potentiometric measurements. This electrode enabled the accurate determination of pH by measuring the potential difference across the glass membrane, which is proportional to the hydrogen ion activity in the solution being measured.

The development of the glass electrode was a significant milestone in the field of pH measurement, as it provided a reliable and practical method for accurately measuring pH in various applications. Prior to this innovation, colorimetric methods using pH indicators were widely used, but they were limited in accuracy and required visual comparison with color charts.

B. Importance and Applications of pH Measurement

The measurement of pH is crucial in various fields, including:

- Environmental monitoring: Monitoring pH levels in water bodies, soil, and air to assess environmental quality and detect potential contamination.
- Industrial processes: Controlling and monitoring pH levels in various industrial processes, such as chemical manufacturing, food and beverage production, wastewater treatment, and pharmaceutical manufacturing.
- Agriculture: Monitoring soil pH to ensure optimal conditions for plant growth and nutrient uptake.
- Healthcare: Measuring pH levels in biological fluids (e.g., blood, urine) for medical diagnostics and treatment monitoring.

C. pH Measurement Principles

Acid-base equilibrium: The dissociation of acids and bases in aqueous solutions, resulting in the formation of hydrogen ions (H+) and hydroxide ions (OH-).

Hydrogen ion activity: The effective concentration of hydrogen ions in a solution, taking into account factors such as ionic strength and activity coefficients.

pH and pOH: The relationship between pH and pOH (the negative logarithm of the hydro

xide ion concentration) in aqueous solutions, where pH+ pOH = 14 at 25°C.

D. Glass Electrode

The most commonly used pH electrode, consisting of a thin glass membrane selectively permeable to hydrogen ions.

The glass electrode is composed of a pH-sensitive glass membrane and an internal reference electrode. The glass membrane develops a potential difference across its surface, which is proportional to the hydrogen ion activity in the solution being measured. This potential difference is measured against a reference electrode, providing a voltage that can be related to the pH value using the Nernst equation.

E. Nernst Equation and its Significance

The Nernst equation relates the potential difference (E) across the glass membrane to the pH of the solution being measured: [10]

$$E = E_0 + (RT/nF) * \ln([H+]) \tag{1}$$

Where:
E_0 : is a constant potential (reference potential).

R : is the universal gas constant.

T : is the absolute temperature.

n : is the number of electrons transferred in the electrode reaction.

F : is the Faraday constant.

[H+] : is the hydrogen ion activity in the solution

This equation forms the basis for calculating the pH value from the measured potential difference across the glass electrode.

F. Factors Affecting PH Measurement Accuracy

- Temperature effects: pH measurements are temperature-dependent, and temperature compensation is often required for accurate readings.
- Electrode aging and drift: Over time, glass electrodes can deteriorate, leading to drift in the measured potential and reduced accuracy.
- Interference from other ions: The presence of certain ions (e.g., sodium, potassium) in the solution can interfere with the pH measurement.
- Sample characteristics: Factors like ionic strength, viscosity, and suspended solids can affect the accuracy of pH measurements.

III. Circuit Design

The system aims to provide accurate pH measurements by integrating a high-impedance pH electrode, buffer amplifier, analog conditioning circuit, microcontroller with ADC(Analog to digital converter), and output display module. The pH electrode measures the potential difference based on hydrogen ion activity, which requires proper interfacing through a high-input resistance buffer. The analog conditioning circuit processes this signal to a suitable range for the microcontroller's ADC. The microcontroller reads the digitized voltage, applies filtering, calculates the pH using the Nernst equation, and displays the result on the output module. Simulations using Proteus software are conducted to verify the circuit designs and optimize component configurations before implementation. The following sections detail the design of each component, simulation results, and overall system integration for the digital pH meter..

A. Block Diagram

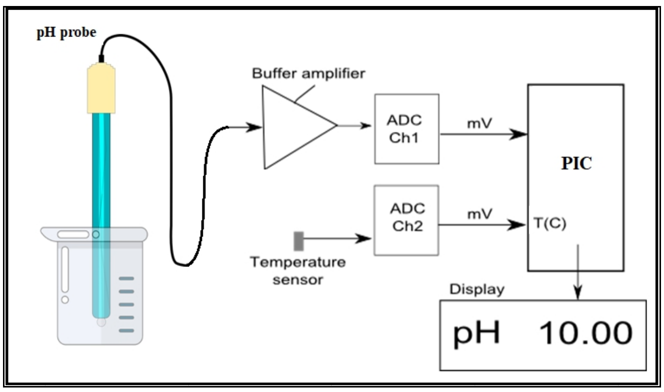

Fig. 2. Simplified block diagram of the digital pH mete

The block diagram in Fig. 2 shows the high-impedance pH electrode measuring potential difference based on hydrogen ion activity. This signal goes through a buffer amplifier and then an ADC (Ch1) to the microcontroller. A temperature sensor is connected to a separate ADC (Ch2) for temperature compensation. The microcontroller calculates the pH using the digitized voltage and temperature, applying filtering algorithms. The calculated pH value is displayed on an output module.

B. *pH probe*

The E201-C BNC pH probe is an economical choice for a reliable pH electrode. Its main parameters are as follows :

TABLE I. PH PROBE PARAMETERS

Parameter	***Range***
Measuring Range	0 to 14 pH
Measuring temperature	0 to 80 C°
Zero point	7 +/- 0.5 PH (25C°)
Internal resistance	250M Ω(25C°)
Percent theoretical slope	(pts) >= 98.5
Response time	< 2minute
Noise	< 0.5 mVEquations

IV. ELECTRICAL CIRCUIT

The pH meter is basically a simple gain/offset circuit with high input impedance. Its purpose is to convert the voltage range of a typical pH probe (between -0.414 at pH 14 and 0.414V at pH 0). The very high input impedance of the pH meter is critical because pH probe has high output impedance, so we must use an even higher input impedance stage for the pH meter. The different stages of the pH meter are:

A. *Input buffer stage*

The circuit input is the BNC connector R1(1) . R1 limits the input current. 4.7MΩ is quite small compared to the impedance of the probe itself which is over 100MΩ. The input buffer amplifier U4 which is the **CA3240E op-Amp** provides the highest input impedance for our pH meter. It's a simple follower with unitary gain (buffer) and it just reproduces the input voltage to its output.

Following the Buffer we find a low pass RC filter (C1 and R2) which prevents higher frequencies to get into our circuit. Its cut-off frequency is set at about 94Hz (1/RC). Anything higher than that will be attenuated. The output of this stage is between -0.414 at pH 14 and 0.414V at pH 0.

B. *Gain Stage circuit*

P1 and the resistors R4 and R3 form the gain circuit. Using op-amp formula, the gain can be adjusted between (R4+R3+P1)/(P1+R3) and (R4+R3+P1)/(R3), which in this case turns out to be between 3.6 and 8. Why this range? A pH probe typically returns between -0.414 and 0.414V, and we'd like to have +/-2V at the output of this stage. Thus we need a gain of 5.

C. *Offset stage*

This op-amp changes the previous stage's +/-2V output range into a 0 to 4V range. This stage also needs to invert the signal, since with 2V input , pH is 0 and with -2V, pH 14. The op-amp is used in a inverting summing amplifier configuration. The pH signal coming through R5 will see a negative gain of -R8/R5. we add the offset which is obtained by -5V going through R7, R6 and P2. The gain for this branch is: -R8/(R7+P2+R6). The extremes for P2 (0 or 10KΩ) give an adjustment range of 1.6V to 2.5V which will be added to the pH signal. Note that the ideal offset value of 2V sits nicely in the middle of this interval. The resistors R9 and R10 are just for voltage division in case we need to change the range , which depends on the ESP32 internal ADC limitation of 3.3V input. The output of this stage is between 0V at pH 0 and 4V at pH 14 .

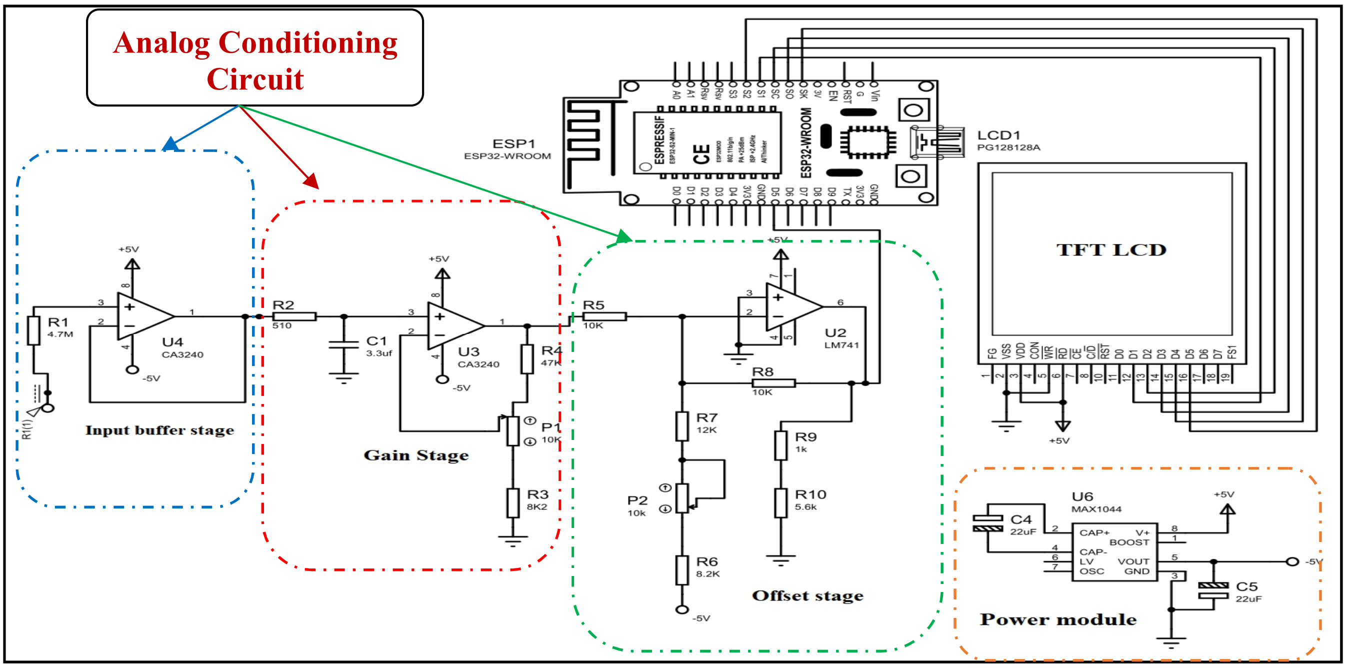

Fig. 3. Electrical circuit of the digital pH mete

D. *ESP32 microcontroller [11]*

The ESP32 is a low-cost, low-power microcontroller from Espressif Systems, featuring a dual-core 32-bit processor, Wi-Fi and Bluetooth connectivity, and a rich set of peripherals. Like the ADCs, capable of measuring signals from sensors and external circuitry.One ADC channel is used to digitize the conditioned pH signal, while the other channel reads the temperature sensor for compensation. The ESP32's processing power and memory resources allow for efficient execution of filtering algorithms, pH calculation using the Nernst equation, and data processing for display.

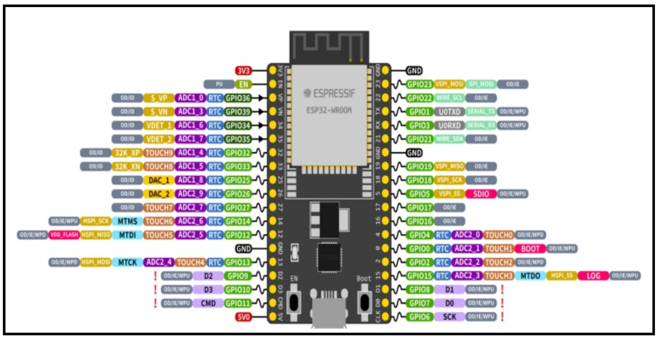

Fig. 4. ESP32 microcontroller pin-out

E. *Power module [12]*

This module provides regulated power supply voltages of 5V and 3.3V, commonly required for powering various electronic components and circuits. The module typically includes a USB connector or a barrel jack for connecting an external power source, such as a USB cable or a wall adapter. It has an on-board voltage regulator that converts the input voltage (usually 5V or higher) to the desired output voltages of 5V and 3.3V.

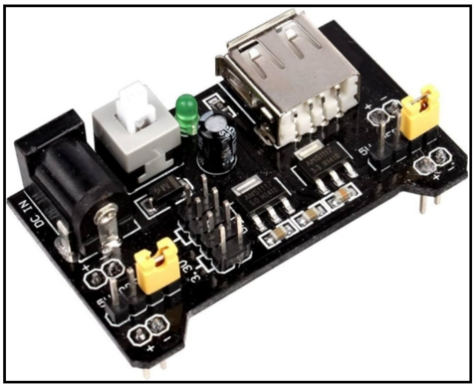

Fig. 5. 5/3.3V Power module

F. *TFT display*

TFT (Thin-Film Transistor) LCD display with a 65K color depth and a resolution of 128x128 pixels was used for the digital pH meter. This type of display offers several advantages over traditional character-based LCD displays like the 16x2 LCD:

- High resolution and color depth: The 128x128 resolution and 65K colors provide a more visually appealing and informative display, allowing for graphical representations, icons, and clearer text rendering.
- Larger display area: The 1.44-inch display size offers a larger viewing area compared to the compact 16x2 LCD, enabling the display of more information simultaneously.
- Graphical user interface (GUI): The TFT display supports the creation of graphical user interfaces, which can enhance the user experience and make the pH meter more intuitive to operate.
- Better readability: The high resolution and color depth of the TFT display ensure better readability, especially in varying lighting conditions, making it suitable for both indoor and outdoor applications.

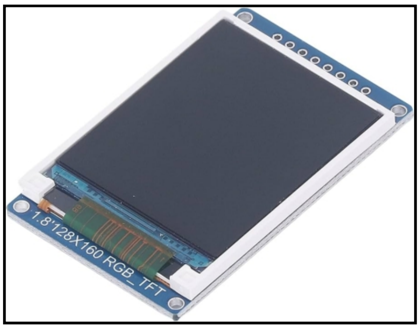

Fig. 6. 1.44 Inch TFT LCD 65K Color 128x128 Display

G. *MAX1044 Voltage converter*

The MAX1044 is a versatile CMOS voltage converter, primarily used for inverting DC voltage to generate a negative voltage.The MAX1044 operates with input voltages ranging from 1.5V to 10V and can provide a negative output voltage close to the magnitude of the input voltage. It features low quiescent current, low power consumption, and can operate at frequencies up to 10 kHz, making it suitable for battery-powered and low-power applications.

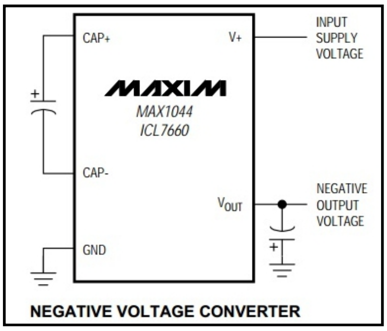

Fig. 7. MAX1044 Internal Circuit

H. *Display stage*

The display stage of the pH meter circuit involves the precise calculation and visualization ofpH values. Initially, the output voltage of the circuit, ranging from 0 to 3.3V, is connected to theADC of the ESP32 microcontroller. The ESP32 then processes this analog voltage, convertingit to digital data. Utilizing the appropriate formulas, such as the Nernst equation, the ESP32calculates the corresponding pH value. This calculated pH value is subsequently displayed on aTFT display interfaced with the ESP32, providing a clear and accurate digital readout of the pHmeasurement.

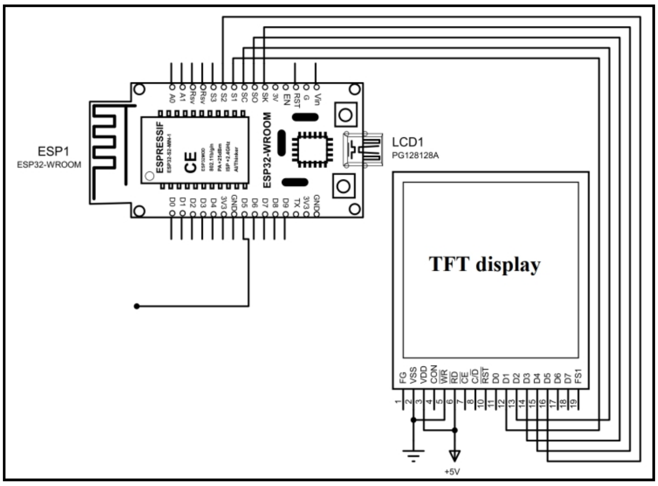

Fig. 8. Display stage

V. CIRCUIT IMPLEMENTATION AND TESTS

A. Implementation

First, the pH probe is placed securely to ensure stability, with its leads easily accessible for connection. The pH probe output is connected to the input of the buffer amplifier, whose output is then wired to the gain stage, and subsequently, the offset stage. The final output from the analog conditioning circuit is connected to one of the ADC pins of the ESP32. The TFT display module is connected to the appropriate GPIO pins on the ESP32 for data and control signals. All connections are double-checked for accuracy, ensuring there are no loose wires, and verifying that the power supply connections are correct.

B. Microcontroller Programming

Programming the ESP32 microcontroller involves writing code to read the ADC values, apply filtering algorithms, and calculate the pH value. The Arduino IDE is used to program the ESP32, with the board package installed and the ESP32 connected to the computer via USB. Necessary libraries for ADC reading and TFT display control are included. The code initializes the ADC pin and sets up the TFT display. The analogRead() function reads the voltage from the ADC pin, and a filtering algorithm, such as a moving average filter, is implemented to smooth out the readings. The Nernst equation is applied to convert the filtered voltage reading to a pH value, adjusted for calibration constants and temperature compensation if needed. The TFT library functions are used to display the calculated pH value on the screen. The code is uploaded to the ESP32, with serial output monitored to verify ADC readings and calculated pH values, ensuring they are as expected, and the pH value is displayed on the TFT screen.

C. Circuit Calibration rules

First we short circuit the input, this is essentially equivalent to having a perfect electrode in a perfect pH7 solution in a perfect world. we Measure the output of the gain stage and confirm that it remains at zero when the gain is modified with P1. Still with the input shorted, set the offset with P2 so that the output of offset stage is 2V. This setting was little difficult as we don't have very accurate material. but we can just set it around 2V,The question is simple: what should be shown by the display for the voltage found after offset stage ? Since 2V is pH 7 the relation is pH=V*7/2.

Given the voltage that we have after offset stage we can thus obtain what we want to see on the display. For example, if the voltage is 2.053V then the value that should be shown on the display is 7.186. Note that we will choose an output value of 3 to 3.5V to obtain a greater display value and thus a greater accuracy in this setting. Now we can just read the value displayed by the TFT.

D. Calibration with known-pH

The pH-meter must now be calibrated with known-pH solutions before it can be used. This should be done before each use. we have a neutral pH solution (pH=7) and a alkali of pH= 9. First we put the pH probe in the pH 7 solution and adjust the offset (P2) to reach pH 7 on the LCD display.

The gain potentiometer (P1) should not have any effect on the reading since any gain applied to a zero voltage will yield a zero output after the gain stage, knowing that a pH=7 corresponds to a probe signal of 0V, If we see a (significant) variation when changing P1 then it means that the pH of our solution is not exactly 7. Note also that the response of the pH meter strongly depends on the probe and the calibration should be performed with stable values only.

After calibrating the offset, we put the probe in the pH=9 solution (gently we must wipe the probe to avoid contamination of the solutions). we won't touch the offset potentiometer P2 anymore, instead we set the gain (P1) so that the display reads pH=9.

The calibration is finished! now we can just use some solutions with known pH values to test the functioning of the pH meter after the calibration process.

If we noticed a gain-variation with P1 during the offset calibration (done with P2) then we will have to repeat the calibration several times; it means that our pH 7 solution was not really pH 7.

This iterative process can diverge if we are not close to the solution. In fact, we could actually use a pH 4 and pH 9 solutions for the calibration, but this would obviously force an iterative calibration process and this is often difficult.

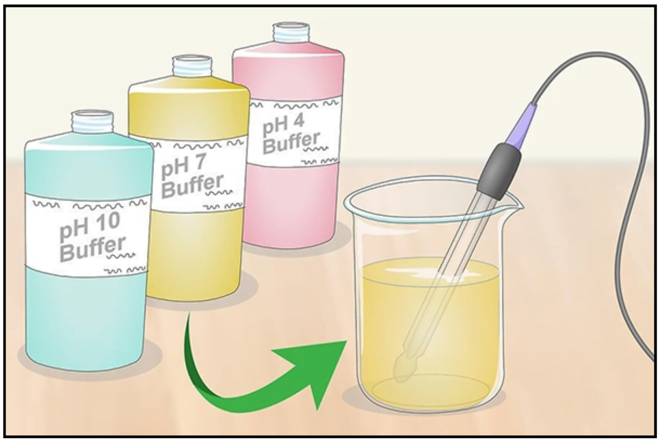

Fig. 9. The Calibration Process

E. Troubleshooting

During implementation, various issues may arise. Common problems include no reading or incorrect reading, which can be addressed by checking all connections on the breadboard to ensure they are secure, verifying the pH probe's functionality, and ensuring the buffer amplifier provides the expected high input impedance.

To test the circuit without pH electrode we just replaced it with a dc voltage of the ranges -0.414V and 0.414V. Table II collect the 3 input voltages as well as the associate pH values recorded on the screen of the TFT display.

TABLE II. TESTS

Input voltage	*The output pH value*
220 mV	3.40
0 mV	7.13
-220 mV	11.14

It's worth noting that, in our case, the final pH value is not based on a single input measurement. Instead, we collect a specified number of voltage readings (approximately 5000) and calculate their average to obtain a more accurate final pH value.

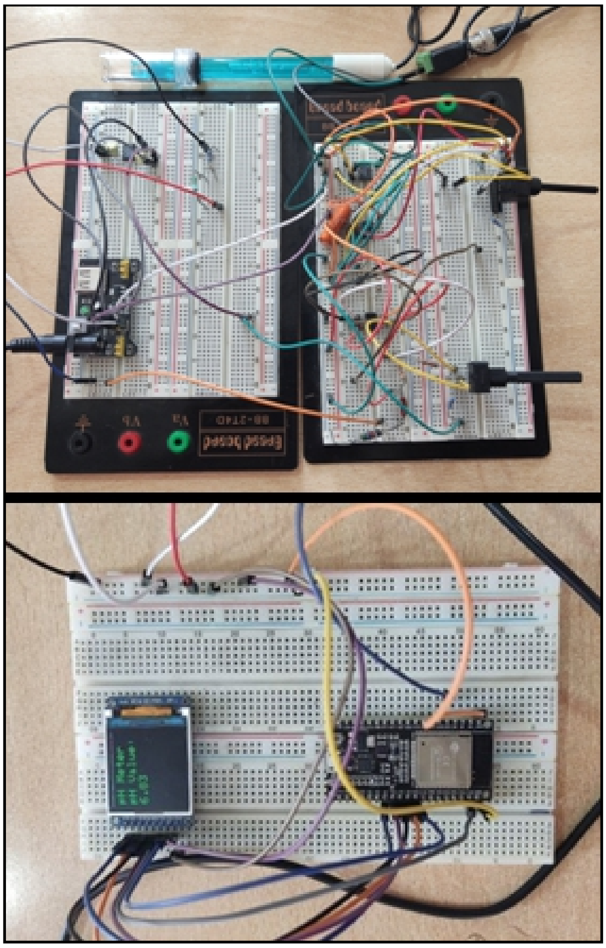

Fig. 10. Circuit implementation

After initial testing and troubleshooting, final adjustments are made. The filtering algorithm is enhanced to improve performance, and the display update routine is optimized to ensure smooth and accurate pH readout, with careful attention to refresh rates and data update intervals.

VI. Conclusion

The digital pH meter has become an indispensable tool in various fields due to its critical role in monitoring and controlling pH levels. We designed and implemented a digital pH meter, achieving highly satisfactory results. The device accurately measures the potential difference from a high impedance glass electrode, conditions the signal, and uses an ESP32 microcontroller to convert and display pH values on a TFT screen. Through meticulous design, using rigorous testing, the system demonstrated high reliability and precision in various pH environments. Although the current design lacks temperature compensation, which is essential for accurate pH measurement across different temperatures, this device successfully applied advanced concepts from analog and digital electronics, reinforcing our theoretical knowledge with practical application. The integration of a microcontroller highlights the modern approach in circuit design. Future enhancements could include adding temperature compensation to improve accuracy and extending the device's functionality to meet broader application requirements. This design not only showcased our ability to create and implement a complex digital system but also provided a solid foundation for further innovations in digital pH measurement.

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